



OPEN Optimized CNN-based ensemble deep learning approach for potato leaf disease detection with data augmentation

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This paper explores the use of optimized convolutional neural networks (CNNs) to classify diseases affecting potato leaves using TensorFlow-2. The dataset, sourced from Kaggle's Plant Village repository, includes 152 images of healthy potato leaves and 1000 images each of early and late blight. The methodology covers data preparation, model architecture design, training, evaluation, and deployment. During data preparation, the data set was split into training sets (80%) and testing sets (20%), with images resized to 128x128 pixels. The Deep Learning (DL) models built using CNN with 4 different optimizers (ADAM, SGD, RMSPROP, and ADAMAX) and trained using a sparse categorical cross-entropy loss function, include multiple convolutional and pooling layers for feature extraction, and fully connected layers for classification. Early stopping was used to prevent overfitting. Model performance was assessed using accuracy, loss curves, confusion matrix, ROC curve, precision recall curve, classification report, and F1 score. In addition, we have used data augmentation to balance the dataset by increasing healthy potato leaves 6 times and the use of Ensemble Deep Learning (EDL). EDL10 which contains DL1 (CNN + ADAM), DL2 (CNN + SGD), DL3 (CNN + RMSPROP) and DL4 (CNN + ADAMX) performs best with a accuracy score of 97.0%. This highlights the importance of data balancing and the use of the ensemble classification approach for the detection of blight in Potato Leaves.

Potatoes (*Solanum tuberosum*) serve as a primary dietary staple for millions globally. However, potato cultivation is significantly threatened by fungal and oomycete diseases, specifically early blight (*Alternaria solani*) and late blight (*Phytophthora infestans*), which severely restrict plant growth. The prevalence of these diseases minimizes agricultural yield and undermines food security by reducing the quality of crops destined for the market. Consequently, efficient and accurate monitoring is essential to optimize disease management and mitigate crop loss. Conventional diagnostic methods, such as farmers visually inspecting crops and conducting specialized laboratory tests, are time-consuming, labor-intensive, and prone to human error or subjective interpretation. The capabilities of a very recent trend in machine learning, the use of a convolutional neural network (CNN), can automate the process of diagnosing illness that is both fast and less expensive. In this study, we present the results of an investigation on the relative effectiveness of CNNs for the classification of potato infections

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by using the dataset from the Plant Village database hosted on Kaggle. Through a systematic analysis of CNN performance across varying hyperparameter configurations, we aim to achieve optimal classification accuracy while maintaining computational efficiency and preventing model overfitting. The objective seeks to enhance the knowledge on CNN accuracy while classifying potato diseases as well as helping to build an effective, flexible, and durable plant disease analysis system. The agricultural sector is a cornerstone of the global economy, providing food security, employment, and revenue. 53.3% of the Indian worker population is engaged in the agriculture sector, which constitutes 18% of the GDP of the country¹. These are patterns that are experienced in other low- and middle-income countries. Notably, the agricultural sector's contribution to India's economy has shown encouraging growth, rising from 17.6% to 20.2% over the past three years^{2,3}, which underscores its significance to the national economy.

Plant diseases and pests are among the most important threats that undermine the quality and productivity of foodstuffs and agriculture. Physical methods are often ineffective in containing widespread outbreaks in most cases. Early detection and a system for close monitoring are imperative in the prevention and maintenance of the quality of crop diseases. Late blight, caused by *P. infestans*, is particularly devastating; in regions like Colombia, it affects numerous potato varieties at high densities, necessitating heavy pesticide use. Traditional leaf inspection methods in these contexts are often hampered by subjective evaluation, high labor costs, and significant time requirements^{7,8}. Patterns resulting from infections and pest destruction are often repeated patterns that can be used for a proper diagnosis⁹. However, manual diagnosis by farmers or non-experts can be inaccurate¹⁰, leading to inappropriate treatments that are detrimental to both the environment and crop quality. These detection problems have received other solutions due to advances in computer imaging. Some imaging techniques have also been used in the study to develop several models that aid in the diagnosis and classification of plant diseases. Such simple approaches corresponding to spatial transformations are still in practice, including the techniques of segmentation and manual feature extraction¹¹. For example, Dubey et al.¹² considered the classification of purely diseased leaf sections and used a multiclass support vector machine (SVM) along with K-means clustering. A group of meteorological and statistical data examined by Yun et al.¹³ allowed the use of a probabilistic neural network (NN). Other models have also been used for the recognition of plants that expose diseases such as SVM, K-mean clustering, and backpropagation NN as reported by Liu et al.¹⁴.

With the growth of artificial intelligence (AI), recent work has focused on machine learning (ML) and deep learning (DL) to improve the accuracy of disease recognition^{15,16}. For example, a study¹⁷ in this regard has tried various CNN designs such as GoogLeNet, AlexNet, and ResNet⁷⁰ which aimed to classify diseases in tomato leaf samples, and found that ResNet was a better performing model. The efficiency of the LeNet architecture in the detection of banana leaf disease was evaluated with gray and color scale images¹⁸. In addition, a study¹⁹ attempted to evaluate five CNN architectures: AlexNet, AlexNetOWTbn, GoogLeNet, Overfeat, and VGG, using state-of-the-art DL models like GoogLeNet, ResNet50, ResNet101, Inception-v3, InceptionResNetv2, SqueezeNet. In²⁰ the Inception-v3 model was applied for the detection of diseases on cassava plants. One of the methods was suggested to duplicate and collect random healthy potato leaf samples to balance the imbalance in the PlantVillage dataset. This study also found that reducing the number of pooling operations relative to convolutional operations improved the performance of the model. By maintaining a ratio of approximately 3:2 for convolution to pooling operations, the model demonstrated improved accuracy in detecting potato blight, effectively minimizing the loss of salient features and improving feature extraction processes.

Key contributions

The proposed Ensemble Deep Learning (EDL) approach addresses existing challenges in image classification by leveraging optimizer-level diversity during CNN training. This results in an effective and efficient solution for detecting blight in potato leaves. The final proposed approach incorporates the use of data balancing through augmentation and use of EDL techniques. We can summarize our contributions in the following manner.

- *Optimizer-Driven CNN Training* We evaluated CNN performance using multiple optimizers (ADAM, SGD, RMSprop, and ADAMAX) to identify distinct convergence patterns for potato disease classification.
- *Ensemble Deep Learning (EDL) Framework* We developed an ensemble architecture that integrates multiple DL models to enhance overall classification robustness and accuracy.
- *Mitigation of Class Imbalance* We addressed the scarcity of healthy leaf images by applying data augmentation, specifically increasing the healthy class representation by sixfold to ensure a balanced training environment.
- *Comparative Performance Analysis* We conducted a rigorous evaluation of model performance before and after data augmentation and ensemble integration to quantify the improvements in diagnostic reliability.
- *Explainable AI (XAI) Validation* We utilized Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the decision-making process of the EDL model, ensuring the predictions are based on relevant biological markers.

Paper organization

The remainder of this paper is organized as follows. Section 2 presents the related work; Sect. 3 explains the methods and materials and data pre-processing and balancing; Sect. 4 describes the proposed architecture; Sect. 5 presents the result and discussion; Sect. 6 outlines the comparative analysis; and Sect. 7 provides the conclusion and future scope.

Related work

Diseases such as early and late blight are among the headwinds of such a critical food crop as potatoes and, if not managed, can lead to significant yield losses. Most of the existing studies in this area are based on the observation of crops by agronomists, which is often time-consuming and prone to human error. In recent years, convolutional

neural networks (CNNs) have emerged as powerful tools for automating the detection and classification of plant diseases^{21–24,69}. Rayhan Asif et al.²⁵ developed a CNN-based method to accurately recognize and categorize disease patterns in potato leaves, focusing on reliable discrimination between healthy and infected samples. Similarly, Rozaqi and Sunyoto²⁶ examined automated disease diagnosis using CNN algorithms, demonstrating the advantages of deep learning models in agricultural applications. Bangari et al.²⁷ provided a comprehensive overview of CNN-based disease detection techniques, emphasizing the role of advanced machine learning in improving agricultural productivity. Lee et al.²⁸ proposed a deep learning-based fabrication method for identifying wellness defects in potato leaves, while Agarwal et al.²⁹ focused on optimizing CNN parameters and architectures for timely potato disease diagnosis. Rashid et al.³⁰ introduced a multilayer deep learning framework aimed at comprehensive potato leaf disease management.

Khobragade et al.³¹ highlighted the importance of accurate disease diagnosis using CNN-based methods to enhance crop yield, whereas Baranwal et al.² investigated deep learning models for potato disease classification. These works, together with earlier reviews^{31–36}, further established CNNs as essential tools for addressing complex agricultural challenges. Mahum et al.³⁴ demonstrated the effectiveness of efficient deep learning models for potato leaf disease identification in precision agriculture, while Islam et al. explored image segmentation combined with multiclass SVMs for disease classification. Joseph et al.³⁷ laid the groundwork for CNN-based early and late blight detection systems, and Sharma et al.³⁸ discussed the growing adoption of deep learning techniques for plant disease diagnosis. Although Slot et al.¹⁶ conducted an extensive comparative study and Asfaw³⁹ emphasized the importance of parameter tuning, optimal configurations for potato disease diagnosis were not clearly identified. Chu et al.⁴⁰ proposed AI based image reconstruction model. Liu et. al.⁴¹ proposed image regulation method.

Beyond potato-specific studies, recent works on other crops further demonstrate the effectiveness of hybrid and lightweight deep learning frameworks for plant disease identification. For instance, a hybrid framework combining modified state-of-the-art CNN models (VGG16, Xception, and ResNet152V2) with feature selection and traditional machine learning classifiers was proposed for tea leaf disease identification⁵⁵. By using deep CNNs as feature extractors and selecting the most discriminative features, the approach achieved a remarkable average accuracy of 99.5% with an AUC of 1.0, highlighting the benefit of hybrid CNN–ML pipelines in agricultural disease recognition.

Similarly, a lightweight and resource-efficient approach was introduced for cashew nut and fruit disease detection using a custom small-sized CNN (CAS-CNN) and a modified MobileNetV2 (CAS-MODMOBNET)⁵⁶. With extensive data augmentation and architectural optimization, the CAS-MODMOBNET model achieved an accuracy of 99.8% and an AUC of 1.0, demonstrating that compact CNN architectures can deliver state-of-the-art performance while maintaining low computational complexity, which is particularly suitable for real-world agricultural deployment.

In light of the growing interest in CNN-based methods for detecting potato diseases, this review highlights key contributions that advance agricultural technology and support improved crop management. CNNs efficiently extract complex spatial features from plant images and consistently outperform traditional machine learning techniques in disease classification tasks. For example, Sharma et al.⁴² developed a CNN-based predictive model for classifying paddy plant diseases using image processing techniques. Table 1 summarizes the related work on potato blight detection.

Materials and methods

In this paper, we implement a Convolutional Neural Network (CNN) architecture⁷¹ using Tensor Flow 2 for the classification of potato diseases. The data set, obtained from the Kaggle Plant Village⁴³, consists of 1000 images of each early and late blight, together with 152 images of healthy potato leaves, showing that the data are highly imbalanced. In this paper, the following steps are carried out to carry out the experiments.

- 1. *Data Preparation* The dataset was split into 80% for training and 20% for testing. From the training portion, 10% was left further for validation. To ensure consistent input size and efficient processing, all images were resized to 128×128 pixels. A sample of the data set is illustrated in Fig. 1.

References	Methodology	Gap	Dataset	Volume	Analysis
44	VGG19, DenseNet169, NASNet, ResNet50V2, InceptionV3, Xception	Limited disease range	Kaggle	1500	Good performance but lacks diversity and interpretability
45	Ensemble: MobileNet, ResNet, Inception	No focus on explainability or real-world tests	Public	10,000	Accurate ensemble, lacks XAI and deployment insights
46	MDSCIRNet with multi-head attention	No real-time or varied dataset testing	Public	5500	Promising hybrid model, needs larger validation
47	DL-based ensemble for 4 diseases	Only 4 types tested	Custom	4200	Emphasizes XAI, but needs broader disease scope
48	ResNet50, DenseNet201	No real-world application focus	PlantVillage	6000	Strong ensemble, lacks deployment context
49	Krill Herd-optimized CNN	No healthy leaves, only 2 diseases	Custom	2500	Novel method, needs more disease types
50	CNN	Highly imbalanced data	Custom	3200	Data imbalance needs resolution
51	CNN-LSTM	Limited diseases, lacks interpretability	Standard	4800	Effective combo, test with diverse data
52	Two-stage approach	Narrow disease scope	Public	5000	Structured model, needs broader coverage

Table 1. Comparison of Methods for Potato Leaf Disease Detection.

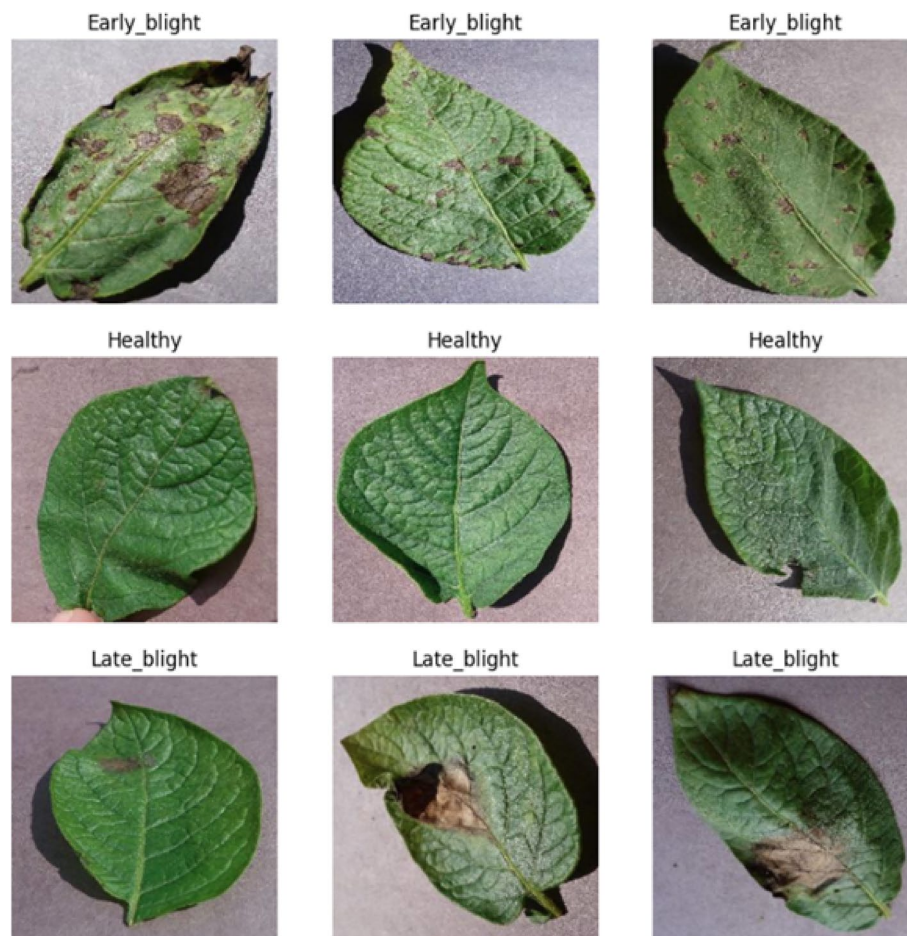


Fig. 1. Sample Images of Potato Plant Disease.

2. *System Architecture* Several convolutional and pooling layers were used in the CNN model architecture⁵⁸ to extract features, and fully connected layers were used for classification, as shown in Fig. 2. The model architecture included input, rescaling, convolutional, max-pooling, flattening, dense, and dropout layers.
3. *Model Training* The sparse categorical cross-entropy loss function (for integer labels) was used to train the model. Early stopping was used to avoid overfitting. The model was trained using four different optimizers (ADAM, SGD, RMSPROP, and ADAMAX) to compare their performance. An ensemble of various CNN models based on different optimizers was used to construct the final blight detection model.
4. *Model Evaluation* The trained model's performance was evaluated using a range of metrics, including training and testing accuracy, loss curves, confusion matrices, ROC curves, precision-recall curves, and the F1 score.
5. *Data Balancing* The unbalanced data set was balanced using a data augmentation technique. In this process, the number of images of healthy potato leaves increased sixfold. Subsequently, the training and evaluation steps were repeated in the balanced data set. Finally, results were compared between the unbalanced and balanced versions.

System architecture layers

This section gives an overview of the various layers of model architecture used in this work.

1. *Input Layer* The input layer receives the raw image data and defines its shape in terms of height, width, and number of color channels, as represented in Eq. 1.

$$X \in \mathbb{R}^{H_i \times W_i \times C} \quad (1)$$

where: X = Input tensor H_i = Height of the image in pixels W_i = Width of the image in pixels C = Number of color channels (e.g., 3 for RGB)

2. *Preprocessing Layer* This layer prepares the input data by applying transformations that help the model learn efficiently. Specifically, a rescaling operation is applied to normalize pixel values from the range [0, 255] to [0, 1], thereby reducing computational cost and improving learning⁵⁹. This is represented in Eq. 2.

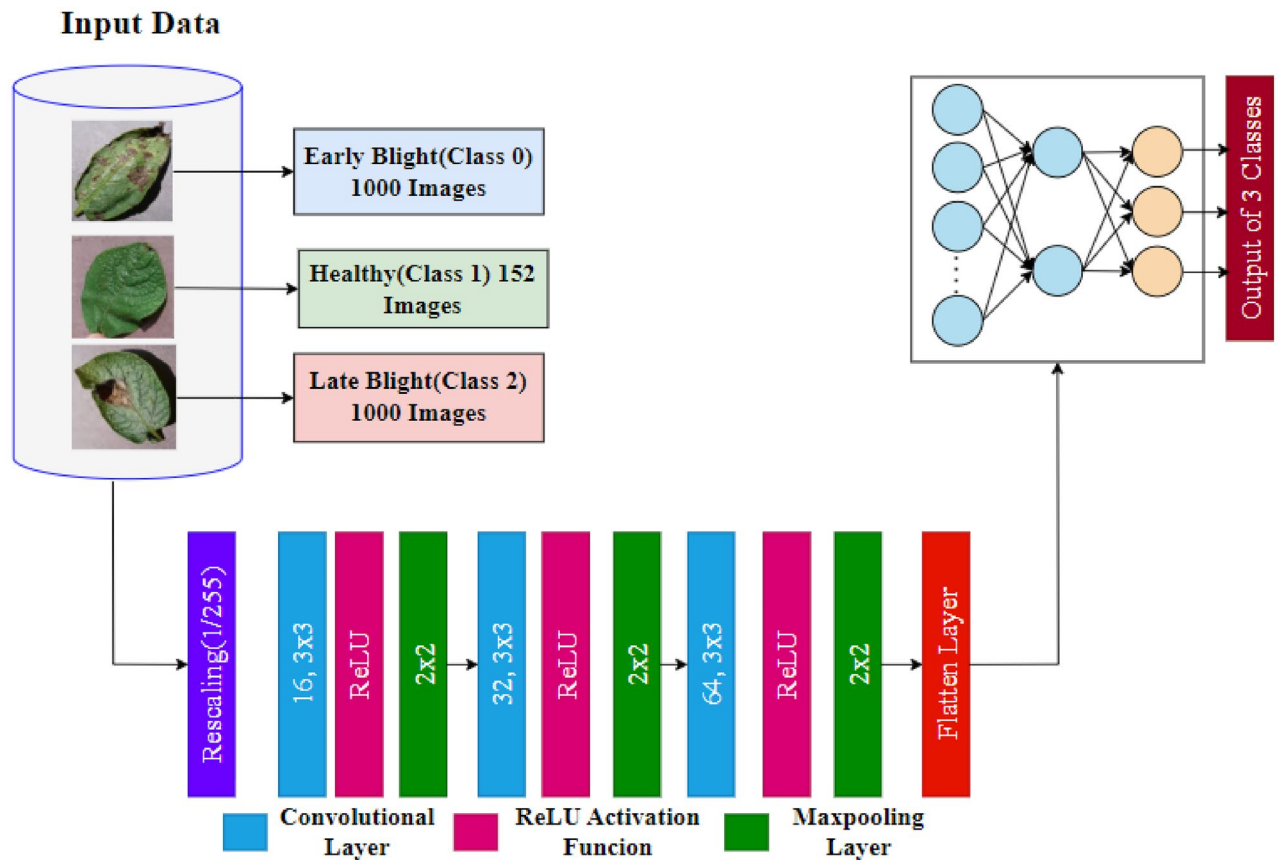


Fig. 2. Architecture of CNN Model used to Classify without Data Augmentation.

$$X_{rs} = \frac{X}{255} \quad (2)$$

where: X = Original pixel value (0–255) X_{rs} = Rescaled pixel value (0–1)

- Convolutional Layer** This layer extracts features from the input using a set of learnable filters (kernels). It detects important patterns such as edges, textures, and shapes⁶⁰. In our model, each convolutional layer uses 16 filters of size 3×3, followed by a ReLU activation function⁵⁷ to introduce non-linearity, and a max pooling operation to reduce spatial dimensions. The output of the convolution operation for each filter f is given by Eq. 3.

$$O_f(i, j) = \sum_{m=0}^{k_h-1} \sum_{n=0}^{k_w-1} \sum_{c=0}^{C-1} X(i+m, j+n, c) \cdot K_f(m, n, c) + b_f \quad (3)$$

where: F = Number of filters (16) (i, j) = Spatial coordinates of the output feature map $K_f(m, n, c)$ = Value of filter f at position (m, n) for channel c b_f = Bias term for filter f

- Flatten Layer** The flatten layer transforms the multi-dimensional output of the convolutional layers into a one-dimensional vector to prepare it for the dense (fully connected) layers⁶¹. This vector $\vec{v}$ is formed as shown in Eq. 4.

$$\vec{v} = [X(0, 0, 0), X(0, 0, 1), \dots, X(0, 0, C-1), \\ X(0, 1, 0), \dots, X(H-1, W-1, C-1)] \quad (4)$$

- Fully Connected Layer** Also known as dense layers, these layers connect every neuron from the previous layer to every neuron in the current one. They are typically used in the final stages of the model for classification⁶². To prevent overfitting, dropout regularization is applied. The softmax function computes class probabilities. The steps are mathematically defined in Eqs. 5–8.

Class	Number of Images
Early Blight (0)	1000
Healthy (1)	152
Late Blight (2)	1000
Total	2152

Table 2. Distribution of Potato Leaf Disease Before Balancing.

Parameter	Value
Rotation Range	40
Horizontal Shift Range	0.2
Vertical Shift Range	0.2
Shear Range	0.2
Zoom Range	0.2
Horizontal Flip	True
Fill Mode	Nearest
Augmentations per Image	6

Table 3. Data Augmentation Parameters for Balancing the Healthy Class.

$$\vec{y} = W \cdot \vec{x} + \vec{b} \quad (5)$$

$$\vec{y}' = \vec{y} \odot \vec{m} \quad (6)$$

$$\vec{y}_{\text{inference}} = \vec{y}' \cdot p \quad (7)$$

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^m e^{z_j}} \quad (8)$$

where: $\vec{x}$ = Input vector of dimension n W = Weight matrix of shape (m, n) $\vec{b}$ = Bias vector $\vec{y}$ = Output of the dense layer $\vec{m}$ = Dropout mask $\vec{y}'$ = Output after applying dropout p = Dropout rate $\vec{y}_{\text{inference}}$ = Output during inference (scaled) z_i = Logit (score) for class i e^{z_i} = Exponential score, enhances higher scores for classification

Optimizers used

- *Adam (Adaptive Moment Estimation)* Adam combines the advantages of two popular optimization techniques, AdaGrad and RMSprop. Calculate adaptive learning rates for each parameter and incorporate momentum to accelerate convergence and effectively handle sparse gradients⁶³.
- *RMSprop (Root Mean Square Propagation)* RMSprop adapts the learning rate for each parameter individually, helping stabilize the training process. It is particularly effective in nonstationary objectives and is widely used in recurrent neural networks (RNNs) due to its ability to mitigate the vanishing gradient problem⁶⁴.
- *SGD (Stochastic Gradient Descent)* SGD is a simple and efficient optimization technique, especially when combined with momentum. However, it often requires careful tuning of the learning rate and other hyperparameters. Compared to adaptive optimizers, SGD may require more manual effort to achieve optimal performance⁶⁵.
- *Adamax* Adamax is a variant of Adam designed to handle scenarios with large gradient magnitudes. It is particularly effective for dealing with sparse or noisy gradients, offering stable convergence in such cases⁶⁶.

Data balancing and augmentation

This study uses the publicly available PlantVillage dataset, which contains images of potato leaves classified into three classes. Early Blight, Healthy, and Late Blight. Each image has a resolution of 256×256 pixels. The corresponding class labels are encoded as follows: 0 for Early Blight, 1 for Healthy, and 2 for Late Blight. The class-wise distribution of the images in the original data set is presented in Table 2.

As shown, the data set suffers from a significant class imbalance, with Healthy leaves being underrepresented. To address this, a data augmentation strategy was applied to increase the number of images in the Healthy class, thereby balancing the data set⁶⁷. Table 3 outlines the augmentation parameters used to synthetically generate additional images. Each original image from the Healthy class was augmented six times using a combination of transformations such as rotation, shifting, shearing, and flipping. Algorithm 1 shows the complete steps used in balancing the dataset using the augmentation approach. After data augmentation, the number of healthy leaf images increased from 152 to 911, which balances the data as shown in Table 4. Figure 3 shows the architecture of the data augmentation model.

```
1: Input: Original Image  $I \in \mathbb{R}^{H \times W \times C}$ 
2: Output: Set of  $N$  Augmented Images  $\{I_{\text{aug}}^1, I_{\text{aug}}^2, \dots, I_{\text{aug}}^N\}$ 
3: Initialize Augmentation Parameters:
4:   Rotation Range  $R = 40^\circ$ 
5:   Horizontal Shift Range  $H_{\text{shift}} = 0.2$ 
6:   Vertical Shift Range  $V_{\text{shift}} = 0.2$ 
7:   Shear Range  $T_R = 0.2$ 
8:   Zoom Range  $Z_R = 0.2$ 
9:   Horizontal Flip Enabled
10:  Fill Mode = Nearest
11:  Number of Augmented Images  $N = 6$ 
12: for each index  $k = 1$  to  $N$  do
13:   Sample random angle  $\theta_k \in [-R, R]$ 
14:   Sample horizontal shift  $X_k \in [-H_{\text{shift}}, H_{\text{shift}}]$ 
15:   Sample vertical shift  $Y_k \in [-V_{\text{shift}}, V_{\text{shift}}]$ 
16:   Sample shear angle  $\phi_k \in [-T_R, T_R]$ 
17:   Sample zoom factor  $Z_k \in [1 - Z_R, 1 + Z_R]$ 
18:   Sample flip probability  $p_k \in [0, 1]$ 
19:    $I_1 \leftarrow \text{Rotate}(I, \theta_k)$ 
20:    $I_2 \leftarrow \text{Shift}(I_1, X_k, Y_k)$ 
21:    $I_3 \leftarrow \text{Shear}(I_2, \phi_k)$ 
22:    $I_4 \leftarrow \text{Zoom}(I_3, Z_k)$ 
23:   if  $p_k < 0.5$  then
24:      $I_5 \leftarrow \text{FlipHorizontal}(I_4)$ 
25:   else
26:      $I_5 \leftarrow I_4$ 
27:   end if
28:    $I_{\text{aug}}^k \leftarrow \text{ApplyFillMode}(I_5, \text{Nearest})$ 
29: end for
30: Return:  $\{I_{\text{aug}}^1, I_{\text{aug}}^2, \dots, I_{\text{aug}}^N\}$ 
```

Algorithm 1. Data Augmentation Process

Proposed model

This paper proposes an optimized Ensemble Deep Learning (EDL) CNN model that combines multiple Convolutional Neural Networks (CNNs), each trained using a different optimizer. The aim is to improve the accuracy and robustness of potato leaf blight detection. The classification pipeline includes the following stages: data collection, class balancing through augmentation, dataset splitting (80:20), model training, and testing using the proposed EDL architecture. Initially, all images are normalized to a range between 0 and 1. The data set is then divided so that 80% is used for training (with 10% reserved for validation), and 20% is used for testing. Since the data set is imbalanced, especially with fewer images of healthy leaves, data augmentation is applied to the underrepresented class to balance the data set prior to training. While ensemble learning commonly leverages architectural diversity combining different CNN architectures such as ResNet, VGG, or MobileNet—our approach focuses on optimizer diversity within a consistent CNN architecture. This choice is motivated by the theoretical understanding that different optimizers explore the loss landscape differently, converging to distinct local minima even when the network architecture and training data remain the same. For example, adaptive optimizers like ADAM and ADAMAX adjust learning rates based on first and second moments of gradients, while SGD follows a more uniform update rule that can lead to flatter minima with better

Class	Number of Images
Late Blight (0)	1000
Early Blight (1)	1000
Healthy (2)	911
Total	2911

Table 4. Distribution of Potato Leaf Disease after Balancing using Augmentation.

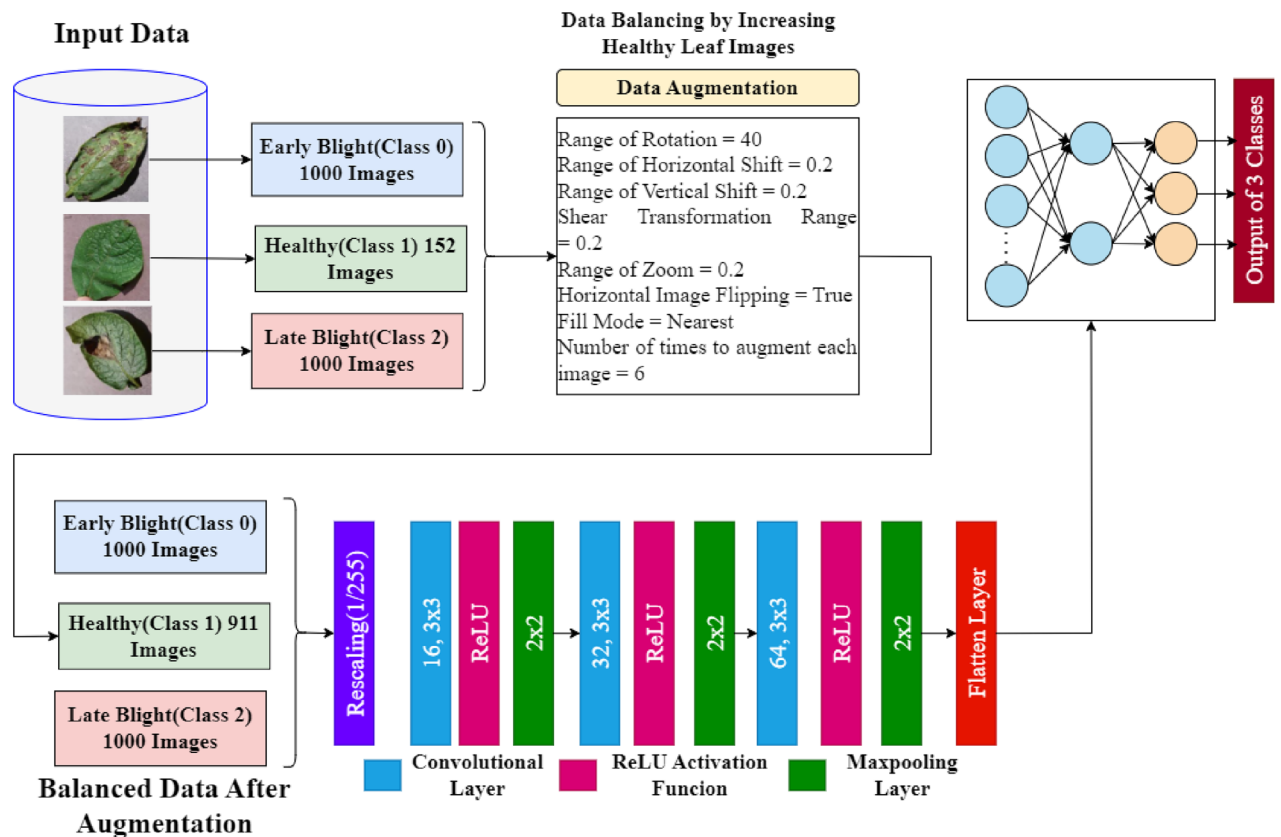


Fig. 3. Potato Leaf Disease Classification Model Architecture with Data Augmentation.

generalization. By ensembling models trained with diverse optimizers, the proposed EDL10 model benefits from reduced correlated errors and improved robustness without the computational overhead of training multiple heterogeneous architectures. A key advantage of the proposed optimizer based ensemble over architectural ensembles is parameter consistency. Using a single optimized CNN backbone ensures that the feature extraction capability is standardized across the ensemble, while the diversity is introduced solely through the weight update physics. This approach is more suitable for resource constrained agricultural IoT devices, as it avoids the memory overhead of storing multiple different architectural frameworks (e.g., VGG's large parameter count vs. MobileNet's depth-wise layers) while still capturing the benefits of ensemble variance. Maintaining a single CNN architecture simplifies model storage and inference pipelines, which is particularly beneficial for real-time deployment in the field.

The aggregation of the individual model outputs is performed using a soft voting mechanism. Let the class probability vector for the i -th base model be $p_i = [p_{i1}, p_{i2}, \dots, p_{iC}]$, where C represents the target classes. The ensemble output p_{ens} is calculated as the arithmetic mean of the probability vectors from the M constituent models:

$$p_{\text{ens}} = \frac{1}{M} \sum_{i=1}^M p_i \quad (1)$$

The final predicted class $\hat{y}$ is determined by selecting the index with the maximum average probability: $\hat{y} = \arg \max(p_{\text{ens}})$. The superior performance of the EDL10 combination (DL1+DL2+DL3+DL4) is attributed to the low error correlation between the optimizers. While ADAM and RMSPROP may share similar error patterns due to their adaptive nature, the inclusion of SGD and ADAMAX introduces diverse decision boundaries. This diversity ensures that the collective decision compensates for individual model weaknesses, leading to the highest observed accuracy.

The proposed architecture as shown in Fig. 4 comprises four individual CNN models trained with the following optimizers: ADAM, SGD, RMSPROP and ADAMAX. These individual models are then combined into ten different EDL configurations for a performance comparison in detecting blight in potato leaves. The CNN architecture used in each model consists of two primary blocks. Each block contains two convolutional layers followed by a max pooling layer. The ReLU activation function is applied after each convolutional operation to introduce nonlinearity and mitigate the vanishing gradient problem. The ReLU function is mathematically defined as

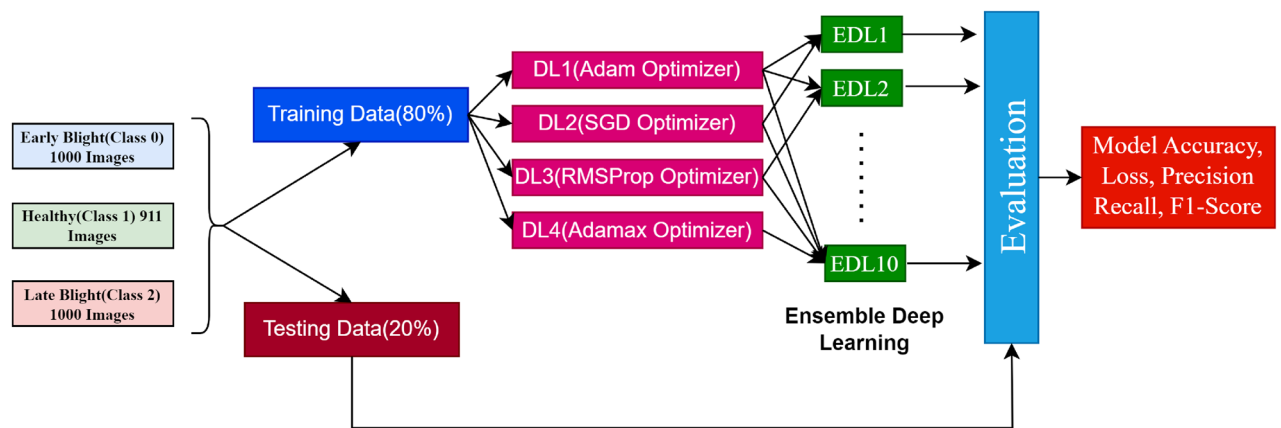


Fig. 4. Architecture of the Proposed Optimized Ensemble Deep Learning Model.

Hyperparameter	Value
Learning Rate	0.0001
Batch Size	32
Kernel/Filter Size	3 × 3
Dropout Rate	0.5
Activation Function	ReLU
Number of Epochs	20
Dense Layer	1 Dense Layer with 128 Neurons
Color Channels	3 (RGB)
Early Stopping	Min Delta = 0.0001, Patience = 5

Table 5. Hyperparameters Used for Training the CNN Model.

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad (9)$$

During training, the input images are resized to $256 \times 256 \times 3$. The first convolutional layer applies 16 filters of size 3×3 , followed by a 2×2 max pooling layer. Subsequent layers increase the number of filters to 32 and then 64, with similar activation and pooling operations. After the convolutional blocks, a flatten layer transforms the 2D feature maps into a 1D feature vector. A dense layer with 128 neurons and ReLU activation is then used to learn higher level representations. Finally, a softmax activated output layer is used to classify the image into one of the three target classes by converting the final output into class probabilities.

Results

This section presents the experimental setup, evaluation metrics, performance analysis, and comparative discussion of the proposed model against existing methods.

Experimental setup

The proposed model was trained and evaluated on the Kaggle platform using the TensorFlow framework (version 2.6.4). A total of 2,652 potato leaf images were used in the experiment. The data set was balanced using data augmentation, containing images in three classes. Early Blight, Late Blight, and Healthy. Table 5 summarizes the hyperparameters used to train CNN based deep learning models.

Evaluation metrics and performance analysis

The performance of the proposed model was evaluated using standard classification evaluation metrics, including accuracy, precision, recall, and F1 score⁶⁸. A confusion matrix was also utilized to visualize the class-wise predictions and assess the strengths and misclassifications of the model. Table 6 defines the formulas used for each metric in the evaluation process.

Results on imbalanced dataset

This section provides an in-depth analysis of the results obtained from various Deep Learning (DL) and Ensemble Deep Learning (EDL) models when applied to the imbalanced dataset. Table 7 presents the evaluation metrics for each model and Fig. 5 offers a visual comparison of their performance. Additionally, Figs. 6, 7, 8

Evaluation Metric	Formula
Accuracy	$A = \frac{TP+TN}{TP+TN+FP+FN}$
Precision	$P = \frac{TP}{TP+FP}$
Recall	$R = \frac{TP}{TP+FN}$
F1 Score	$F1 = 2 \times \frac{P \times R}{P+R}$

Table 6. Evaluation Metrics Used for Performance Analysis.

Model	Accuracy	Precision	Recall	F1 Score
DL1	0.89	0.81	0.90	0.84
DL2	0.87	0.92	0.67	0.69
DL3	0.91	0.87	0.89	0.88
DL4	0.93	0.92	0.89	0.91
EDL1	0.93	0.86	0.92	0.88
EDL2	0.92	0.86	0.91	0.88
EDL3	0.92	0.86	0.91	0.88
EDL4	0.93	0.91	0.90	0.90
EDL5	0.95	0.95	0.89	0.92
EDL6	0.94	0.92	0.91	0.92
EDL7	0.92	0.87	0.91	0.89
EDL8	0.94	0.93	0.93	0.93
EDL9	0.94	0.92	0.90	0.91
EDL10	0.95	0.92	0.92	0.92

Table 7. Evaluation Metrics Comparison on Imbalanced Dataset.

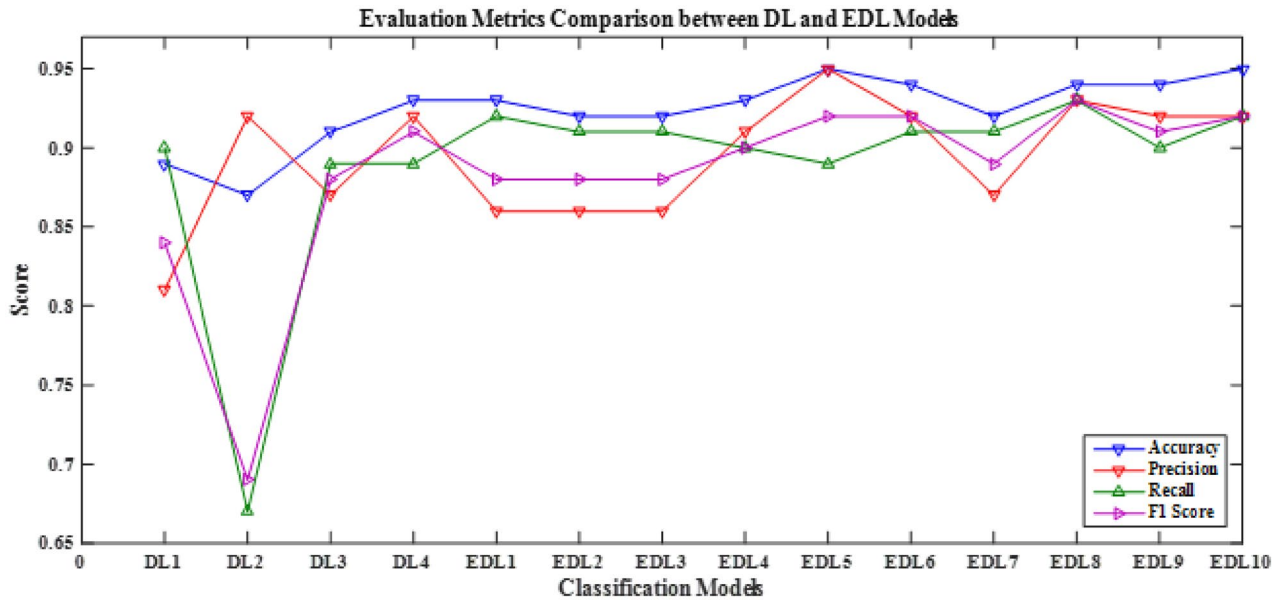


Fig. 5. Comparison of Evaluation Metrics for DL and EDL Models on Imbalanced Dataset.

and 9 illustrate detailed evaluation curves for each DL model (DL1 to DL4), including accuracy, loss, precision-recall, and ROC.

As observed in Table 7, DL4 shows the best performance among individual models with a precision of 0.92. Among all ensemble configurations, EDL5 and EDL10 achieve the highest accuracy of 0.95, showing an improvement over the standalone DL models. EDL5 also achieves the highest precision of 0.95, while EDL8 outperforms others in both the recall and the F1 score with values of 0.93. These results confirm that EDL

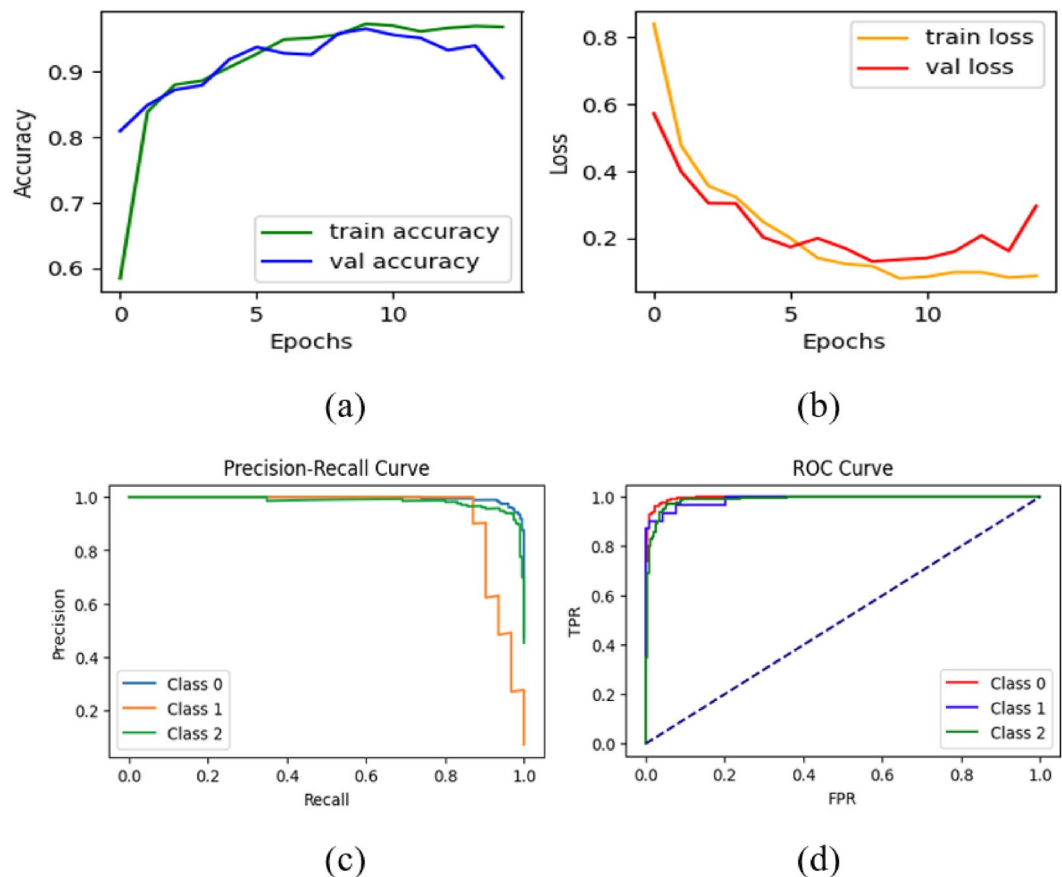


Fig. 6. DL1: (a) Accuracy vs. Epoch, (b) Loss vs. Epoch, (c) Precision-Recall Curve, (d) ROC Curve on Imbalanced Dataset.

models offer improved and more stable performance compared to individual DL models, especially in scenarios involving imbalanced data.

Figures 6, 7, 8 and 9 illustrate training and validation accuracy and loss across epochs. The accuracy trends show a progressive improvement, reaching around 0.9. The loss curves indicate effective learning, although some fluctuations in validation loss appear in later epochs. The precision recall curves reveal high performance for Class 0 and Class 2, while Class 1 (healthy leaves) exhibits reduced precision at higher recall levels. This suggests that the model identifies Class 1 less reliably compared to the others, likely due to the low number of samples in that class. ROC curves demonstrate high true positive rates for all classes, with Class 0 and Class 2 performing slightly better than Class 1, indicating better separability for those classes. These results highlight the robustness of the model in identifying diseased leaves, while also suggesting that performance for the underrepresented class (Class 1) can benefit from further enhancement. In summary, the ensemble models outperform individual CNNs in all evaluation metrics, and EDL5, EDL8, and EDL10 provide particularly strong results. The class imbalance affects the precision and recall for Class 1, confirming the need for balancing strategies or class-specific weighting during training.

Results on balanced dataset using augmentation

This section presents the experimental results after addressing the class imbalance through data augmentation. The effect of augmentation on each Deep Learning (DL) and Ensemble Deep Learning (EDL) model is evaluated using standard metrics including Accuracy, Precision, Recall, and F1 Score, as well as the more robust statistical metrics Cohen's Kappa and Matthews Correlation Coefficient (MCC). Figures 11, 12, 13 and 14 illustrate accuracy, loss, precision-recall, and ROC curves for DL1 to DL4, respectively. Following augmentation, voting ensemble classifiers were used to further enhance model performance. Table 8 summarizes the comparative performance of all models, including the newly added Kappa and MCC metrics. Among the models, EDL10 achieved the highest accuracy of 97%, with corresponding Cohen's Kappa and MCC values of 0.95, outperforming other configurations and demonstrating strong model reliability and agreement beyond chance. Comparative plots of accuracy, precision, recall, and F1 score for all models—before and after balancing—are presented in Figs. 15, 16, 17 and 18. The analysis reveals that all models, except DL2, exhibited significant improvements in all metrics after data augmentation. DL2 displayed consistently poor performance, indicating a tendency to overfit or remain biased towards the dominant classes even after balancing. Figure 10 shows the comparison of the confusion matrix for DL1-DL4 in both imbalanced and balanced datasets. For Class 1 (healthy leaves),

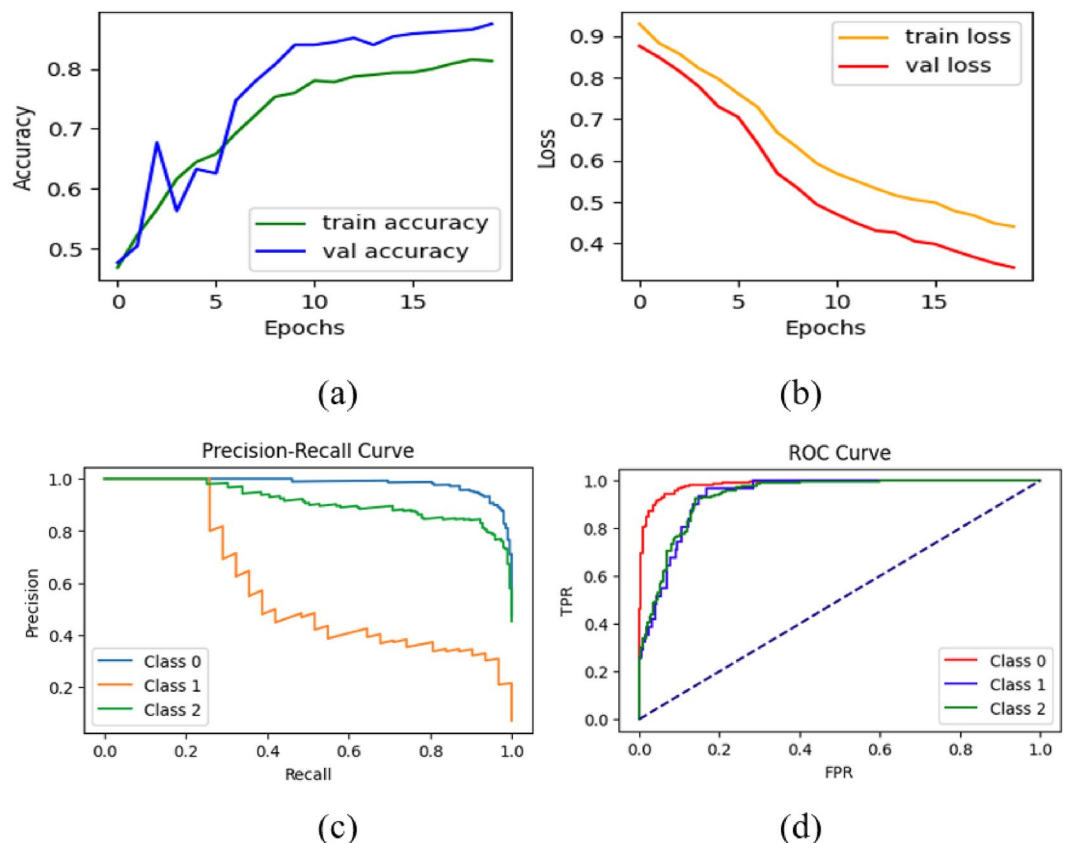


Fig. 7. DL2: (a) Accuracy vs. Epoch, (b) Loss vs. Epoch, (c) Precision-Recall Curve, (d) ROC Curve on Imbalanced Dataset.

which was underrepresented in the original dataset, the mean classification accuracy improved significantly from 67.7% (DL1: 87.1%, DL2: 0%, DL3: 93.5%, DL4: 90.3%) to 84.95% after balancing (DL1: 100%, DL2: 90.4%, DL3: 49.4%, DL4: 100%). This improvement of 17.25 percentage points underscores the effectiveness of data augmentation in improving classification of minority classes (Figs. 12 and 13).

Figures 11, 12, 13 and 14 show clear improvements in training and validation accuracy for DL1, DL3, and DL4, all exceeding 90% accuracy. However, DL2 did not benefit from the augmentation. The loss curves for all models also show successful convergence, with minor fluctuations in validation loss. The precision recall curves indicate that DL1, DL3, and DL4 now classify Class 1 more effectively. However, DL2 continues to struggle, especially in maintaining high precision at higher recall values. ROC curves reaffirm that DL1, DL3, and DL4 achieve high true positive rates in all classes, while DL2 shows suboptimal separation for Class 1.

Figures 15, 16, 17 and 18 provide further insight. DL1 exhibits the highest improvement in all metrics, with the precision increasing by 15.63% and the F1 score by 12.5%. EDL5 shows the least change in precision (1.04%) and recall (7.29%). DL2 demonstrates negative improvement in all metrics, confirming its ineffectiveness even after balancing.

As shown in Table 9, balancing the data set through augmentation significantly improves the performance of the model, particularly for underrepresented classes. DL1 and EDL10 stand out as the best performing models in balanced scenarios, while DL2 remains an outlier requiring further tuning or redesign (Figs. 16 and 17).

XAI results

Gradient-weighted Class Activation Mapping (Grad-CAM) is a widely adopted technique in the domain of Explainable Artificial Intelligence (XAI), offering visual insights into the decision-making process of convolutional neural networks (CNNs). Grad-CAM operates by computing the gradients of the target class with respect to the activations of the final convolutional layer, thereby generating class discriminative heatmaps that highlight regions of the input image most influential in the model prediction. This interpretability is especially critical in vision based applications such as plant disease detection, where understanding the model's focus helps ensure that its decisions align with expert knowledge. Grad-CAM bridges the gap between model internal representations and human interpretability, serving as a tool not only for transparency, but also for model debugging and reliability assessment. Figure 19 presents the Grad-CAM visualizations generated using the best performing ensemble model (EDL10) on sample images from each of the three classes: Early Blight, Healthy, and Late Blight. In the Early Blight sample, the heatmap prominently highlights dark, irregularly shaped lesions areas features typically associated with disease. For the Healthy leaf sample, the visualization shows a more

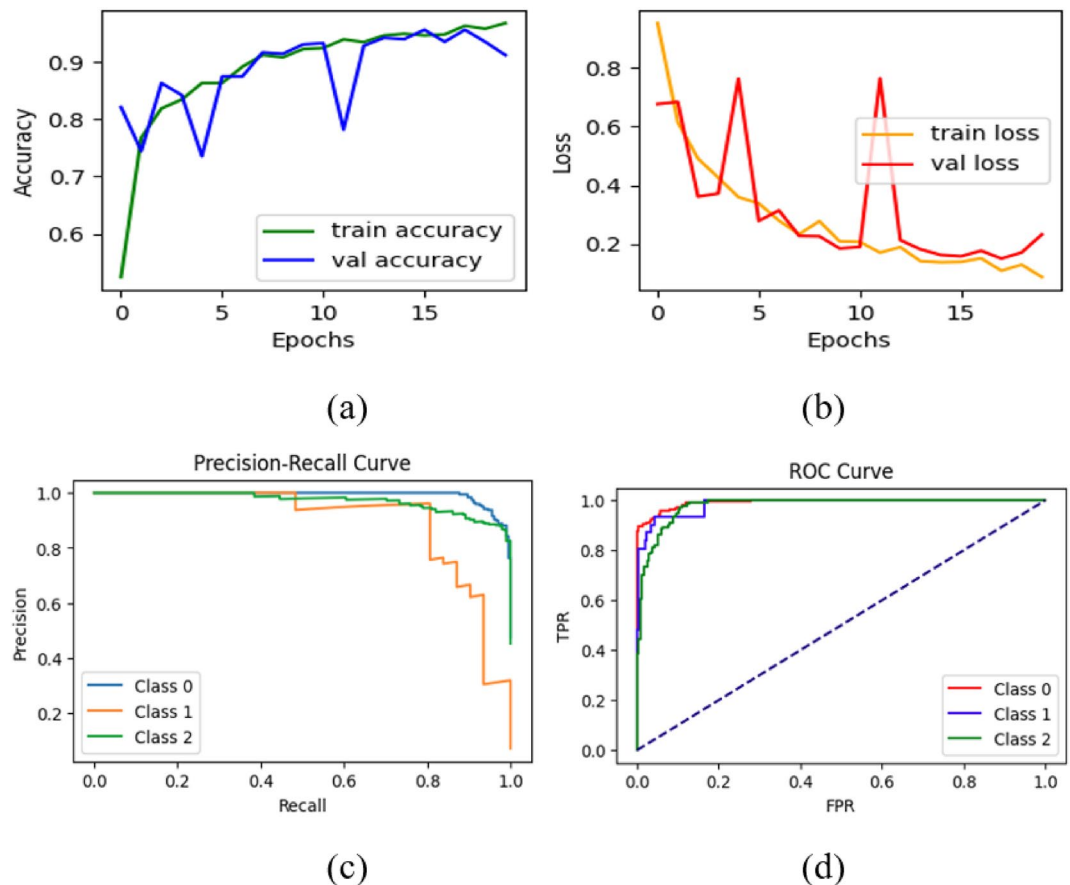


Fig. 8. DL3: (a) Accuracy vs. Epoch, (b) Loss vs. Epoch, (c) Precision-Recall Curve, (d) ROC Curve on Imbalanced Dataset.

distributed activation pattern across the leaf surface. While some localized intensity is observed, further analysis confirms that the model is focusing on the uniform green texture and structural integrity of the leaf rather than background noise or experimental artifacts. This is a critical distinction, as models trained on the PlantVillage dataset can sometimes succumb to “background memorization” bias. In our EDL10 model, the ensemble of different optimizers helps regularize the feature extraction process, ensuring that the network prioritizes the semantic features of the leaf blade over the consistent gray background typical of this dataset. For Late Blight, Grad-CAM emphasizes the brownish and decaying regions of the leaf, consistent with the characteristic symptoms of this condition. The alignment of these heatmaps with the actual physiological symptoms of the potato plants suggests that the model has learned to identify relevant biological markers rather than relying on spurious correlations in the image background. These visualizations demonstrate that the EDL10 model is responsive to areas of semantic significance and disease relevance, thus strengthening trust in its predictive capabilities. By revealing the internal rationale behind the predictions, Grad-CAM supports interpretability and accountability, which are essential for real-world deployment in sensitive domains such as agriculture, healthcare, and environmental monitoring.

Comparison with previous work

To rigorously evaluate the effectiveness of the proposed approach, the performance of the optimized ensemble deep learning model (EDL10) was compared with recent machine learning and deep learning techniques used for potato blight classification. For consistency and fairness in the evaluation, images from the PlantVillage dataset, a widely accepted benchmark in plant disease detection research, were used. Table 10 presents a comparative analysis of the model against previous studies. The results clearly demonstrate that the proposed EDL10 model achieves competitive performance with an accuracy of 97.0%, matching or surpassing the accuracy reported by state-of-the-art methods. Although Singh et al. and Islam et al. achieved relatively high accuracy using traditional machine learning approaches such as SVM with GLCM features, deep learning models such as VGG16, VGG19, ResNet50, and MobileNet reported by Chakraborty et al. show comparatively lower performance. The performance metrics for these baseline architectures were cited from the original studies⁵⁴, which employed standardized training protocols and transfer learning strategies on the PlantVillage dataset. While this study did not perform independent fine-tuning of these architectures, fairness was ensured by comparing against well-established benchmarks reported in the literature. As shown in Table 10, the EDL10 model (97.0%) outperforms several state-of-the-art single architectures like VGG16 and ResNet50. This suggests

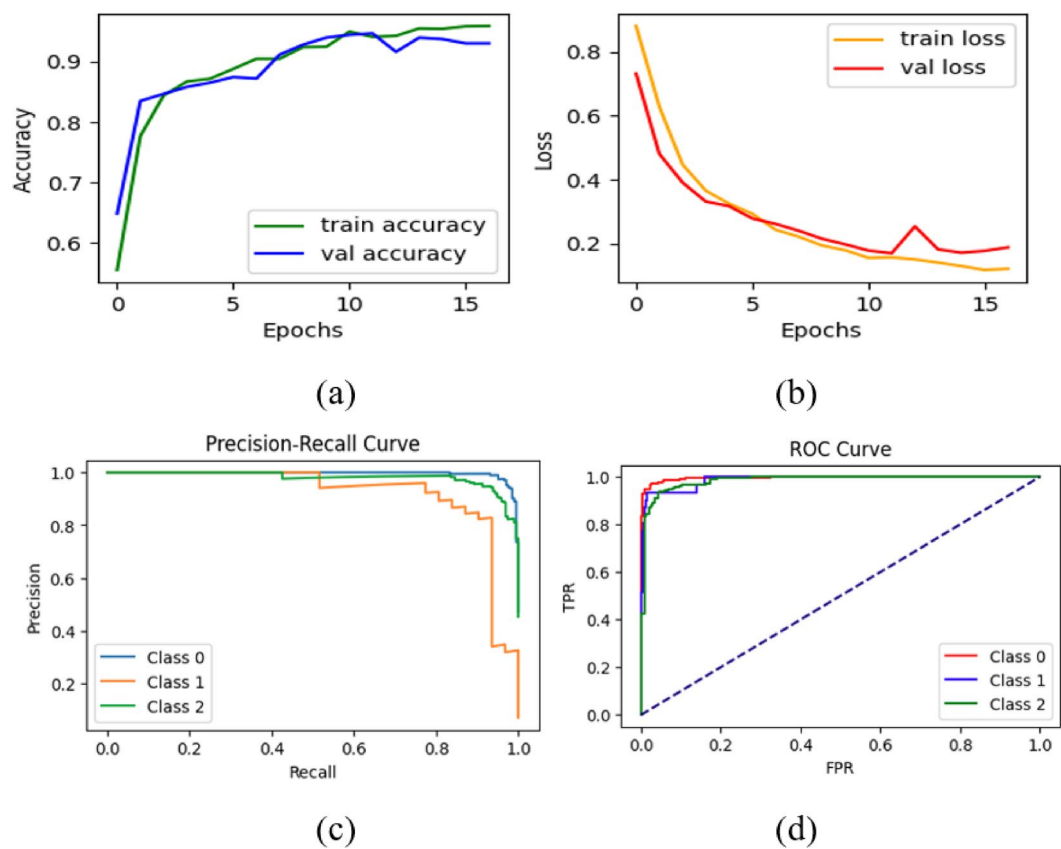


Fig. 9. DL4: (a) Accuracy vs. Epoch, (b) Loss vs. Epoch, (c) Precision-Recall Curve, (d) ROC Curve on Imbalanced Dataset.

Model	Accuracy	Precision	Recall	F1 Score	Kappa	MCC
DL1	0.96	0.96	0.96	0.96	0.94	0.94
DL2	0.56	0.74	0.55	0.52	0.34	0.34
DL3	0.95	0.95	0.95	0.95	0.92	0.92
DL4	0.95	0.95	0.95	0.95	0.92	0.92
EDL1	0.96	0.96	0.97	0.96	0.94	0.94
EDL2	0.96	0.96	0.96	0.96	0.94	0.94
EDL3	0.96	0.96	0.96	0.96	0.94	0.94
EDL4	0.95	0.95	0.95	0.95	0.92	0.92
EDL5	0.96	0.96	0.96	0.96	0.94	0.94
EDL6	0.96	0.96	0.96	0.96	0.94	0.94
EDL7	0.96	0.96	0.96	0.96	0.94	0.94
EDL8	0.96	0.96	0.97	0.96	0.94	0.94
EDL9	0.96	0.96	0.96	0.96	0.94	0.94
EDL10	0.97	0.97	0.97	0.97	0.95	0.95

Table 8. Evaluation Metrics Comparison on Balanced Dataset after Augmentation (including Cohen’s Kappa and MCC).

that for specific tasks like potato blight detection, the variance introduced by optimizer diversity is as effective, if not more so, than simply switching to a deeper or more complex architecture, which often risks overfitting on smaller agricultural datasets. This reinforces the efficacy of the optimized ensemble strategy, which leverages the unique convergence properties of multiple optimizers to create a more robust decision boundary. By maintaining architectural consistency while diversifying the optimization physics, the EDL10 model achieves superior generalization compared to individual high-complexity models that may struggle with the specific visual features of potato leaf lesions.

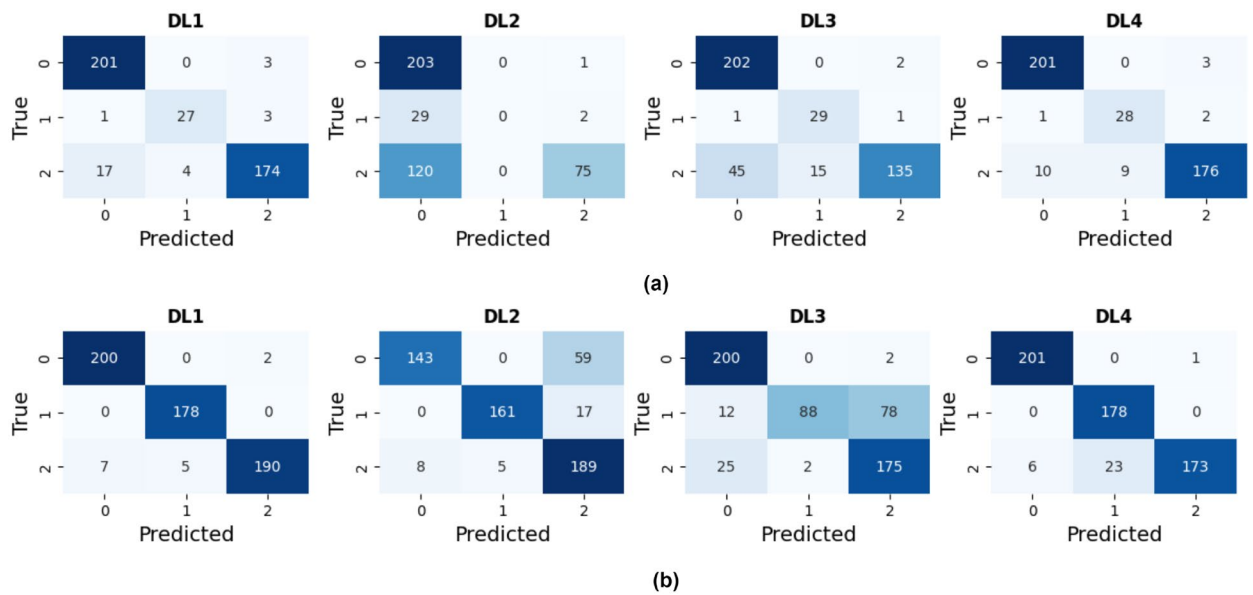


Fig. 10. Confusion Matrix Comparison (a) Imbalanced Dataset, (b) Balanced Dataset.

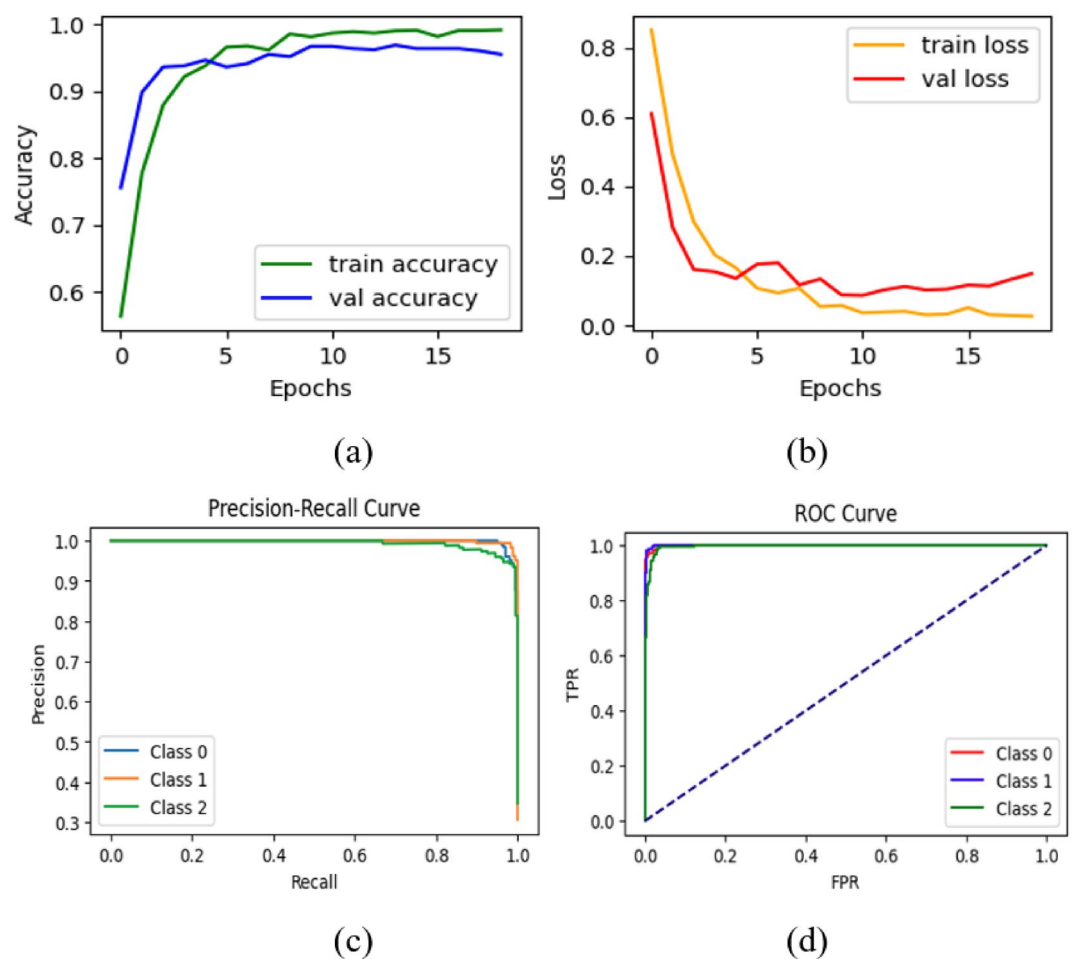


Fig. 11. DL1: (a) Accuracy, (b) Loss, (c) Precision-Recall, (d) ROC on Balanced Dataset.

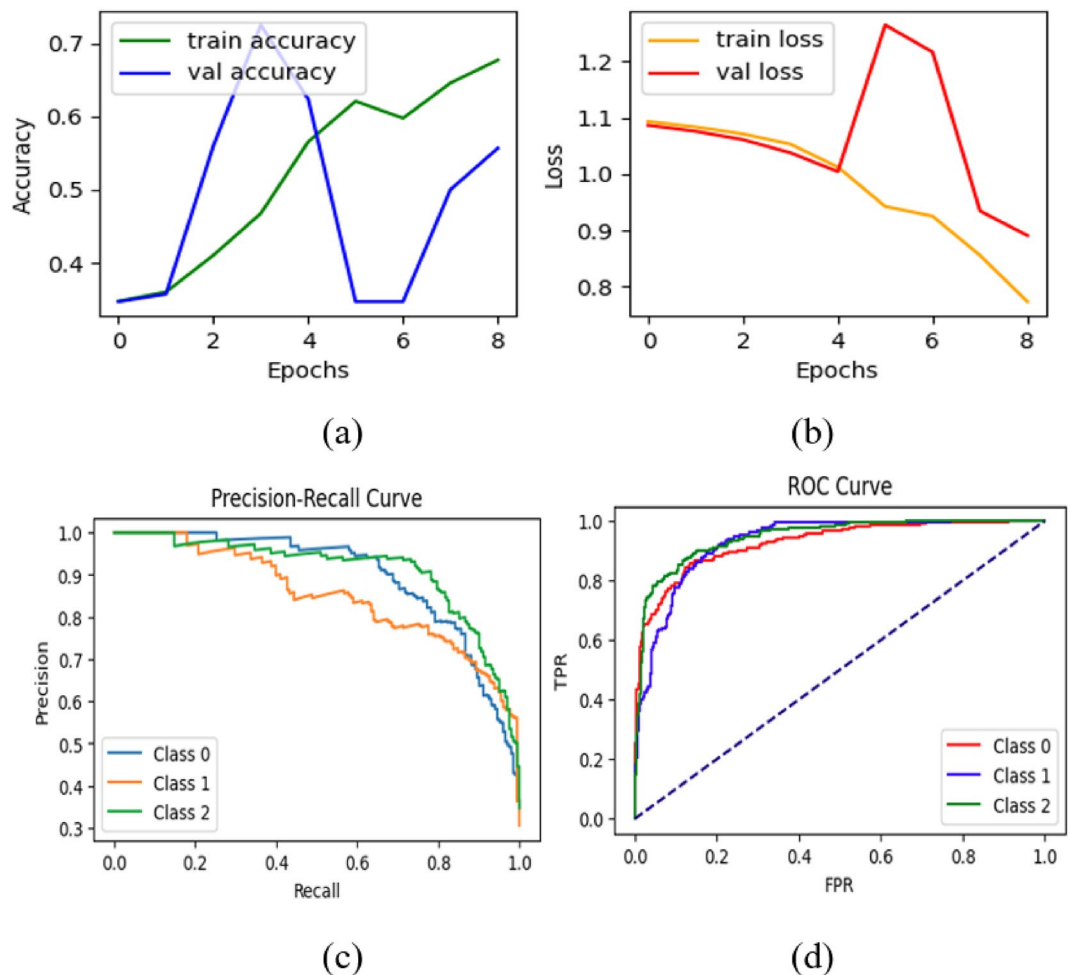


Fig. 12. DL2: (a) Accuracy, (b) Loss, (c) Precision-Recall, (d) ROC on Balanced Dataset.

Conclusion

In this study, we investigated the application of convolutional neural networks (CNNs) to detect blight in potato plants, with a focus on improving model performance through data balancing and optimization strategies. Initially, the models were evaluated using four different optimizers, Adam, SGD, RMSprop, and Adamax, on an imbalanced data set. This phase revealed the critical challenge of accurately identifying the minority class, specifically healthy potato leaves, that were underrepresented in the original data set. To address this problem, data augmentation techniques were used to balance the dataset, resulting in a marked improvement in the model's ability to generalize and accurately classify previously underrepresented classes. This finding underscores the importance of addressing class imbalance, particularly in tasks where minority classes carry crucial diagnostic value. In addition, we explore the effectiveness of ensemble deep learning (EDL) models constructed using combinations of the base CNN models trained with different optimizers. Among these, the EDL10 model demonstrated the highest performance, achieving an accuracy of 97%, thus validating the advantage of the model ensembling for complex classification tasks. In particular, while most models benefited from data augmentation, the DL2 model remained an outlier with limited improvement, suggesting potential areas for future investigation and optimization. In terms of performance gains after balance, DL1 exhibited the highest improvements in accuracy, precision, and F1 score, 7.3%, 15.63%, and 12.5%, respectively. For recall, EDL5 showed the greatest improvement, increasing by 7.29%. Regarding computational complexity, the proposed EDL10 ensemble aggregates four base CNNs, resulting in a total parameter count of approximately 8.48 million. The sequential inference time for the ensemble was recorded at approximately 1.57 seconds for 582 images, which presents a specific computational profile for mobile deployment. Consequently, future work will focus on optimizing this ensemble through knowledge distillation and weight quantization to reduce the computational footprint and latency while maintaining high classification accuracy. These results collectively emphasize the value of balancing data sets and employing ensemble techniques for robust CNN-based classification, particularly in agricultural applications such as identifying healthy versus diseased plant leaves. This work contributes to more accurate and explainable AI-based disease detection frameworks in precision agriculture.

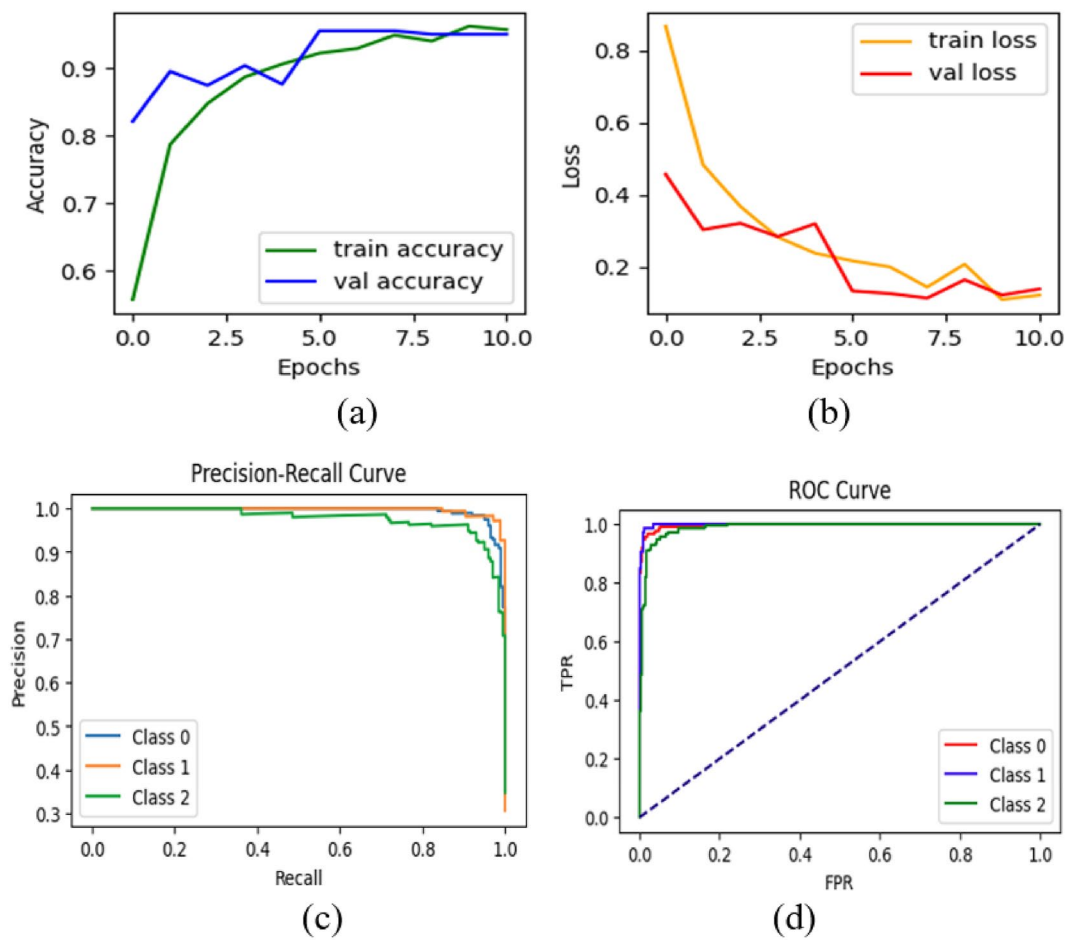


Fig. 13. DL3: (a) Accuracy, (b) Loss, (c) Precision-Recall, (d) ROC on Balanced Dataset.

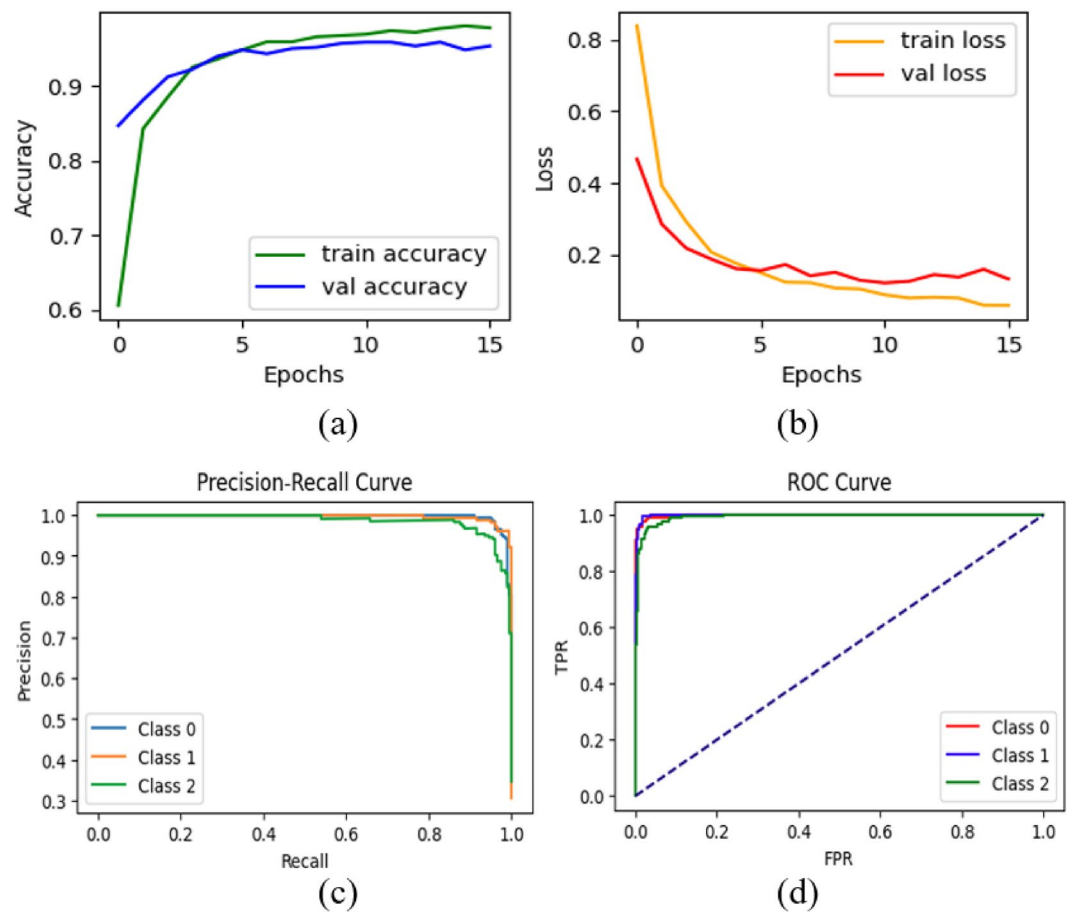


Fig. 14. DL4: (a) Accuracy, (b) Loss, (c) Precision-Recall, (d) ROC on Balanced Dataset.

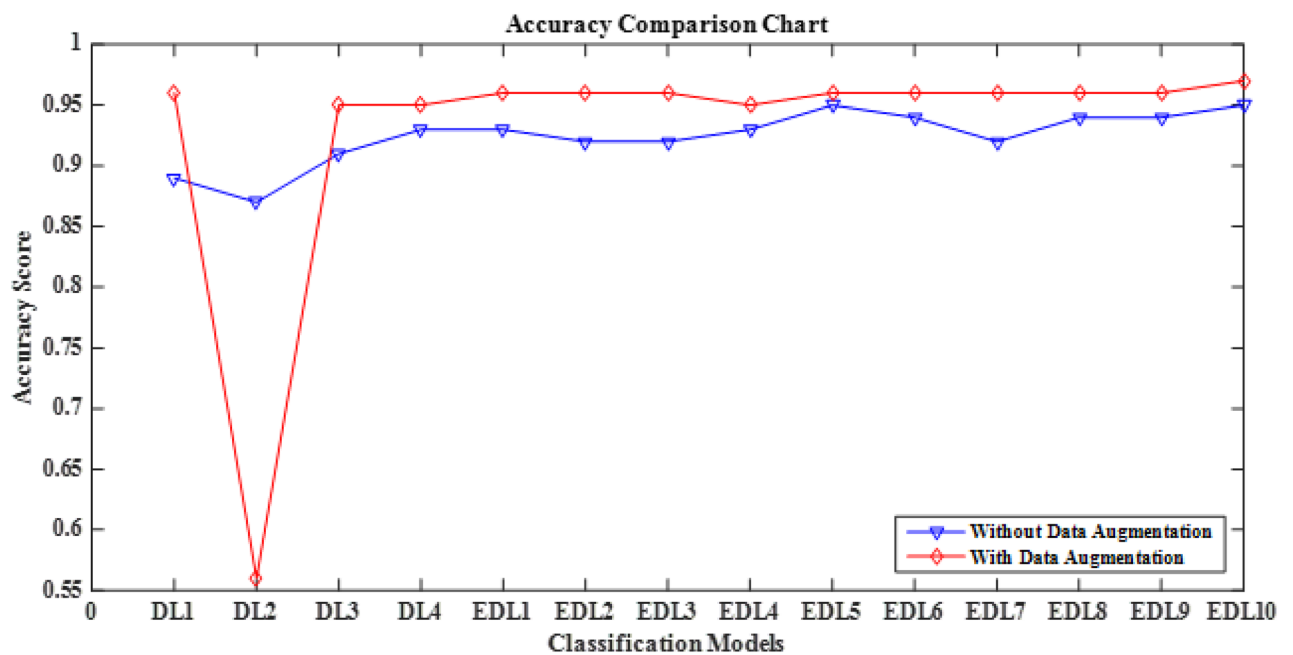


Fig. 15. Accuracy Comparison: Imbalanced vs Balanced Dataset.

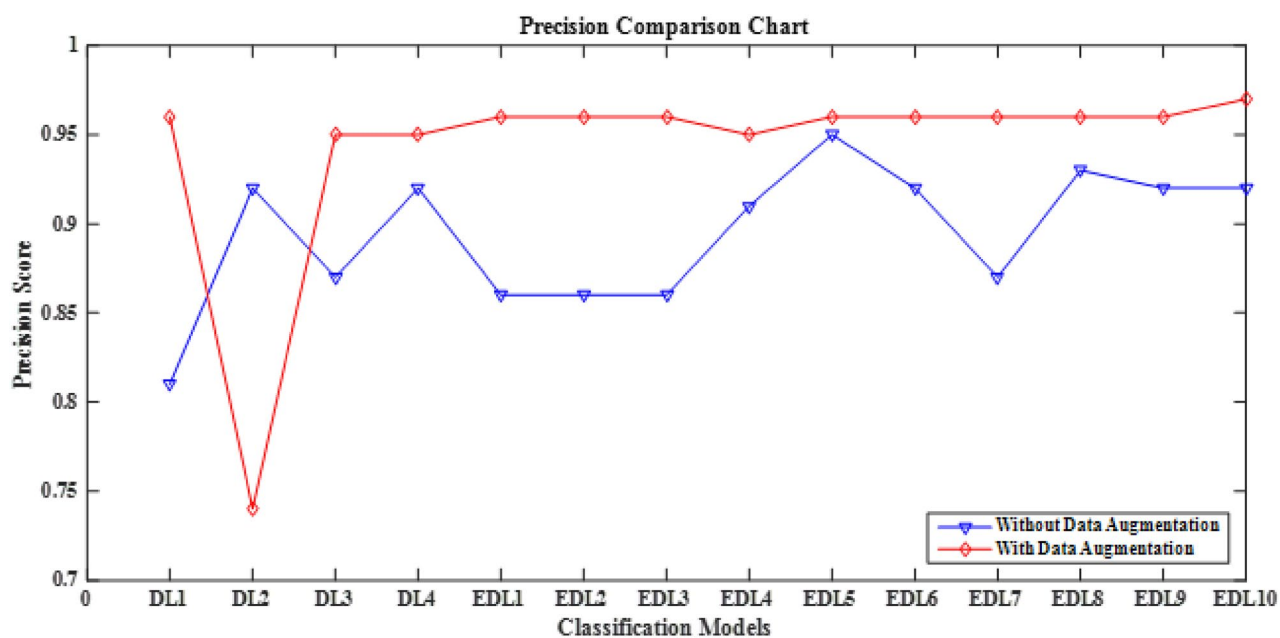


Fig. 16. Precision Comparison: Imbalanced vs Balanced Dataset.

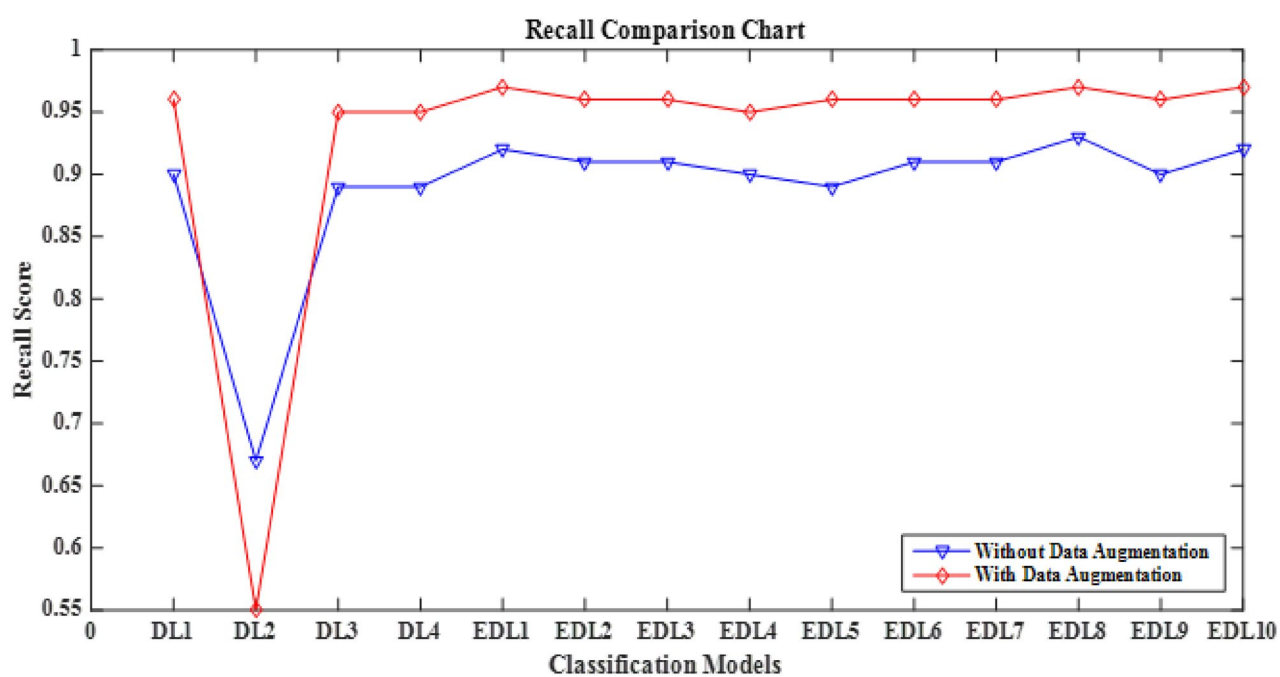


Fig. 17. Recall Comparison: Imbalanced vs Balanced Dataset.

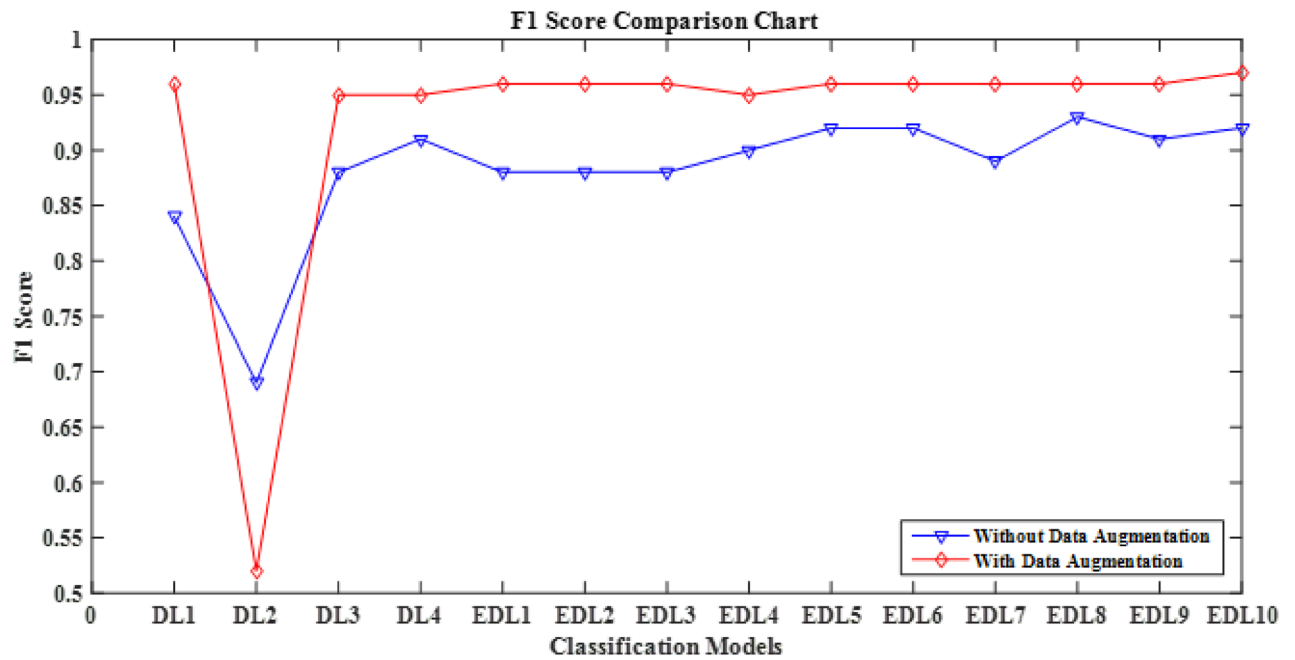


Fig. 18. F1 Score Comparison: Imbalanced vs Balanced Dataset.

Evaluation Metric	Highest % Increase	Average % Increase
Accuracy	7.3% in DL1	3.13%
Precision	15.63% in DL1	6.89%
Recall	7.29% in EDL5	5.61%
F1 Score	12.5% in DL1	6.41%

Table 9. Performance Gains After Data Balancing (Excluding DL2).

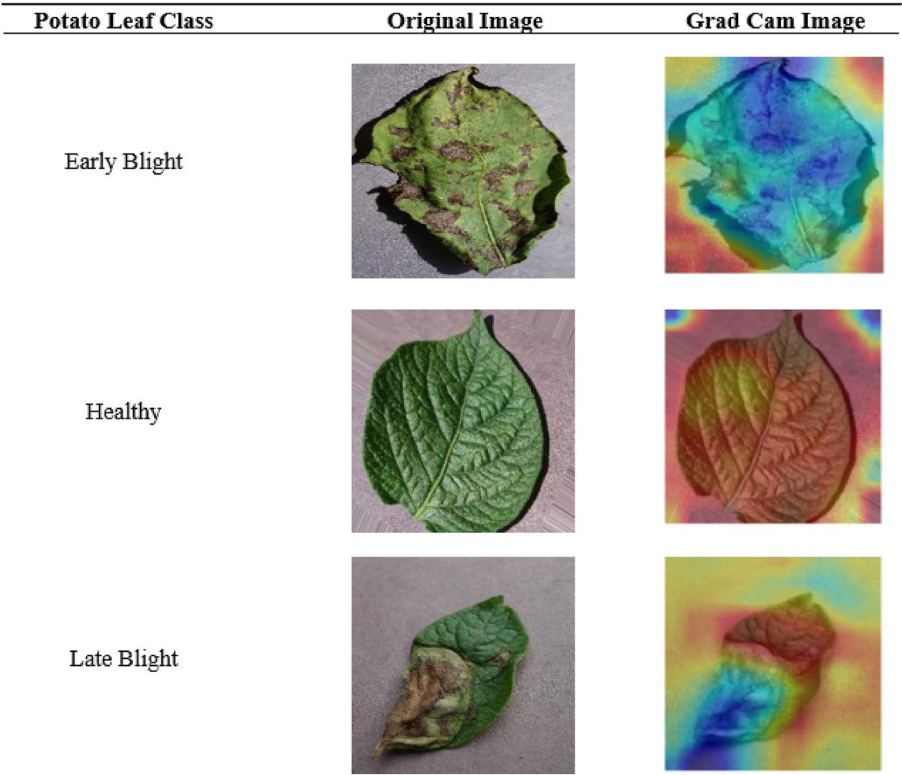


Fig. 19. Grad-CAM Visualizations for EDL10 Model on Early Blight, Healthy, and Late Blight Potato Leaf Samples.

Study	Model	Accuracy (%)
Singh et al. ⁵³	GLCM + SVM	95.99
Islam et al. ³⁶	SVM	95.00
Chakraborty et al. ⁵⁴	VGG16	92.69
	VGG19	80.39
	ResNet50	73.75
	MobileNet	78.84
Proposed (This Study)	Optimized EDL (EDL10)	97.00

Table 10. Comparison with Existing Approaches.

Data availability

The datasets generated and/or analysed during the current study are available at : PlantVillage Dataset, accessed from <https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset>.

Code availability

All data supporting the findings of this study are publicly available. The complete source code is hosted in a DOI-minting repository and has been archived on Zenodo to ensure long-term accessibility and reproducibility. The code is released under an open-source license. The archived version corresponding to this publication is available at : <https://doi.org/10.5281/zenodo.19624017>.

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Declarations

Competing of interests

The authors declare no competing interests.

Additional information

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